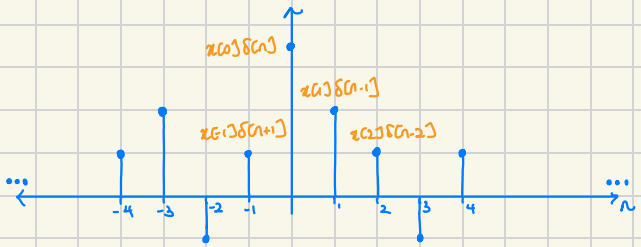
ELG 3125 - Signals & Systems

Lecture 4 - Part 1

Linear Time-Invariant (LTI) Systems

Rep. of Discrete-Time Signals in Terms of Impulses



$$x[n] = \ldots + x[-3]\delta[n+3] + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + \ldots$$

$$\Rightarrow \boxed{x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]} \quad \text{sifting property}$$

Discrete-Time Unit Impulse Response and the Convolution-Sum Representation of LTI Systems

$h_k[n]$: response of LTI system to shifted unit impulse $\delta[n-k]$.

If the linear system is time-invariant $\Rightarrow$ responses to time-shifted unit impulses are all time-shifted versions of the same impulse response:

$$h_k[n] = h[n-k]$$

$$\Rightarrow h_1[n] = h[n-1],\ h_2[n] = h[n-2],\ h_3[n] = h[n-3],\ \ldots,\ h_k[n] = h[n-k]$$

Time-Invariance

$$\delta[n] \rightarrow \boxed{\text{LTI}} \rightarrow h[n] \quad \Rightarrow \quad \delta[n-k] \rightarrow \boxed{\text{LTI}} \rightarrow h[n-k]$$

$$\Rightarrow \quad x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$h[n]$: characterizes the system completely.

$$\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]} \Rightarrow \boxed{y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]}$$

Convolution sum.

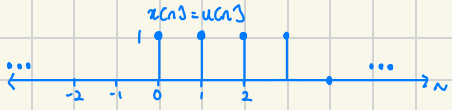
Intuitive way to interpret convolution

If signal B was placed at position t, how much total energy do signal A & signal B share in their overlap?

Represented symbolically: $\boxed{y[n] = x[n] * h[n] = h[n] * x[n]}$

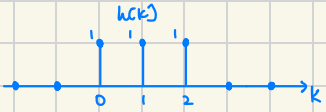
Calculation of Convolution Sum



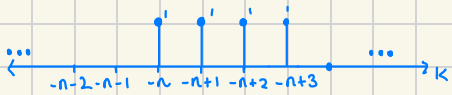


Step 1: Draw $h[k]$ & $x[n-k]$

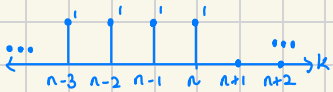
$x[n-k] = x[-k+n]$

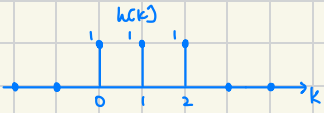


$x[k+n]$

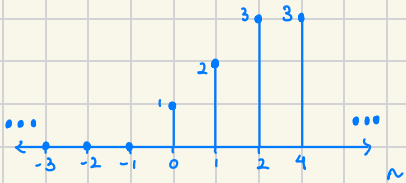


$x[n-k] = x[-k+n]$





Overall convolution graph



$n < 0$: no overlap ⇒ convolution sum = 0 ⇒ $\sum_{k=-\infty}^{n} x[k]h[n-k] = 0$

$0 \le n \le 2$: overlap for non-zero region ⇒ $\sum_{k=0}^{n} x[k]h[n-k] = \sum_{k=0}^{n} 1 = n+1$

$\sum_{k=0}^{n} 1 = n+1$

$n > 2$: $\sum_{k=0}^{2} x[k]h[n-k] = \sum_{k=0}^{2} (1) = 3$

$$y[n] = \begin{cases} 0, & n < 0 \\ n+1, & 0 \le n \le 2 \\ 3, & n > 2 \end{cases}$$